

Task 1 — SQL Benchmark Ground Truth Report

Database: ClassicModels (PostgreSQL) Total Questions: 50 All Queries Verified: ✓

Part 1A — Simple SELECT Queries (Q1–Q20)

Q1: List all products

SQL Query:

```
SELECT * FROM products;
```

Rows Returned: 110

Sample Output:

productCode	productName	productLine	productScale	productVendor	productDescription
S10_1678	1969 Harley Davidson Ultimate Chopper	Motorcycles	1:10	Min Lin Diecast	This replica features working kickstand, front suspension, gear-shift lever, footbrake lever, drive chain, wheels and steering. All parts are particularly delicate due to their precise scale and require special care and attention.
S10_1949	1952 Alpine Renault 1300	Classic Cars	1:10	Classic Metal Creations	Turnable front wheels; steering function; detailed interior; detailed engine; opening hood; opening trunk; opening doors; and detailed chassis.

S10_2016	1996 Moto Guzzi 1100i	Motorcycles	1:10	Highway 66 Mini Classics	Official Moto Guzzi logos and insignias, saddle bags located on side of motorcycle, detailed engine, working steering, working suspension, two leather seats, luggage rack, dual exhaust pipes, small saddle bag located on handle bars, two-tone paint with chrome accents, superior die-cast detail , rotating wheels , working kick stand, diecast metal with plastic parts and baked enamel finish.
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Explanation: Returns all 110 products with full details from the products table.

Q2: Get all customers

SQL Query:

```
SELECT * FROM customers;
```

Rows Returned: 122

Sample Output:

customerNumber	customerName	contactLastName	contactFirstName	phone	address
103	Atelier graphique	Schmitt	Carine	40.32.2555	54, rue Royale
112	Signal Gift Stores	King	Jean	7025551838	8489 S St.
114	Australian Collectors, Co.	Ferguson	Peter	03 9520 4555	636 St Road

Explanation: Returns all 122 customer records including contact info and credit limits.

Q3: Show all orders

SQL Query:

```
SELECT * FROM orders;
```

Rows Returned: 326

Sample Output:

orderNumber	orderDate	requiredDate	shippedDate	status	comments	customerNumbe
10100	2003-01-06	2003-01-13	2003-01-10	Shipped	None	363
10101	2003-01-09	2003-01-18	2003-01-11	Shipped	Check on availability.	128
10102	2003-01-10	2003-01-18	2003-01-14	Shipped	None	181

Explanation: Returns all 326 orders with dates, status, and customer references.

Q4: List all employees

SQL Query:

```
SELECT * FROM employees;
```

Rows Returned: 23

Sample Output:

employeeNumber	lastName	firstName	extension	email	officeCo
1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1
1056	Patterson	Mary	x4611	mpatterso@classicmodelcars.com	1
1076	Firrelli	Jeff	x9273	jfirrelli@classicmodelcars.com	1

Explanation: Returns all 23 employees with job titles and office assignments.

Q5: Get all offices

SQL Query:

```
SELECT * FROM offices;
```

Rows Returned: 7

Sample Output:

officeCode	city	phone	addressLine1	addressLine2	state	country	postalCode	territory
1	San Francisco	+1 650 219 4782	100 Market Street	Suite 300	CA	USA	94080	NA
2	Boston	+1 215 837 0825	1550 Court Place	Suite 102	MA	USA	02107	NA

3	NYC	+ 1 212 555 3000	523 East 53rd apt. 5A Street	NY	USA	10022	NA
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Explanation: Returns all 7 office locations with address and territory info.

Q6: Show all product lines

SQL Query:

```
SELECT * FROM productlines;
```

Rows Returned: 7

Sample Output:

productLine	textDescription	htmlDescription	image
Classic Cars	Attention car enthusiasts: Make your wildest car ownership dreams come true. Whether you are looking for classic muscle cars, dream sports cars or movie-inspired miniatures, you will find great choices in this category. These replicas feature superb attention to detail and craftsmanship and offer features such as working steering system, opening forward compartment, opening rear trunk with removable spare wheel, 4-wheel independent spring suspension, and so on. The models range in size from 1:10 to 1:24 scale and include numerous limited edition and several out-of-production vehicles. All models include a certificate of authenticity from their manufacturers and come fully assembled and ready for display in the home or office.	None	None
Motorcycles	Our motorcycles are state of the art replicas of classic as well as contemporary motorcycle legends such as Harley Davidson, Ducati and Vespa. Models contain stunning details such as official logos, rotating wheels, working kickstand, front suspension, gear-shift lever, footbrake lever, and drive chain. Materials used include diecast and plastic. The models range in size from 1:10 to 1:50 scale and include numerous limited edition and several out-of-production vehicles. All models come fully assembled and ready for display in the home or office. Most include a certificate of authenticity.	None	None

Planes	Unique, diecast airplane and helicopter replicas suitable for collections, as well as home, office or classroom decorations. Models contain stunning details such as official logos and insignias, rotating jet engines and propellers, retractable wheels, and so on. Most come fully assembled and with a certificate of authenticity from their manufacturers.	None	None
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Explanation: Returns all 7 product line categories with descriptions.

Q7: List all payments

SQL Query:

```
SELECT * FROM payments;
```

Rows Returned: 273

Sample Output:

customerNumber	checkNumber	paymentDate	amount
103	HQ336336	2004-10-19	6066.78
103	JM555205	2003-06-05	14571.44
103	OM314933	2004-12-18	1676.14

Explanation: Returns all 273 payment records with amounts and dates.

Q8: Get product names and prices

SQL Query:

```
SELECT "productName", "buyPrice" FROM products;
```

Rows Returned: 110

Sample Output:

productName	buyPrice
1969 Harley Davidson Ultimate Chopper	48.81
1952 Alpine Renault 1300	98.58
1996 Moto Guzzi 1100i	68.99

Explanation: Selects only product name and buy price columns from products.

Q9: Get customer names and cities

SQL Query:

```
SELECT "customerName", city FROM customers;
```

Rows Returned: 122

Sample Output:

customerName	city
Atelier graphique	Nantes
Signal Gift Stores	Las Vegas
Australian Collectors, Co.	Melbourne

Explanation: Selects customer name and city for all 122 customers.

Q10: List employee first and last names

SQL Query:

```
SELECT "firstName", "lastName" FROM employees;
```

Rows Returned: 23

Sample Output:

firstName	lastName
Diane	Murphy
Mary	Patterson
Jeff	Firrelli

Explanation: Returns first and last name of all 23 employees.

Q11: Get all order dates

SQL Query:

```
SELECT "orderNumber", "orderDate" FROM orders;
```

Rows Returned: 326

Sample Output:

orderNumber	orderDate
10100	2003-01-06
10101	2003-01-09
10102	2003-01-10

Explanation: Returns order number and order date for all 326 orders.

Q12: Show product vendor list

SQL Query:

```
SELECT DISTINCT "productVendor" FROM products;
```

Rows Returned: 13

Sample Output:

productVendor
Welly Diecast Productions
Motor City Art Classics
Classic Metal Creations

Explanation: Returns distinct list of product vendors using DISTINCT keyword.

Q13: Get all product codes

SQL Query:

```
SELECT "productCode" FROM products;
```

Rows Returned: 110

Sample Output:

productCode
S10_1678
S10_1949
S10_2016

Explanation: Returns all product codes from the products table.

Q14: List all countries from offices

SQL Query:

```
SELECT DISTINCT country FROM offices;
```

Rows Returned: 5

Sample Output:

country
USA
France
Japan

Explanation: Returns distinct countries where offices are located (5 countries).

Q15: Show all order statuses

SQL Query:

```
SELECT DISTINCT status FROM orders;
```

Rows Returned: 6

Sample Output:

status
Shipped
In Process
Disputed

Explanation: Returns distinct order status values (6 statuses: Shipped, In Process, etc).

Q16: Get all payment amounts

SQL Query:

```
SELECT "customerNumber", amount FROM payments;
```

Rows Returned: 273

Sample Output:

customerNumber	amount
103	6066.78
103	14571.44
103	1676.14

Explanation: Returns customer number and payment amount for all 273 payments.

Q17: List all job titles

SQL Query:

```
SELECT DISTINCT "jobTitle" FROM employees;
```

Rows Returned: 7

Sample Output:

jobTitle

VP Sales
Sales Manager (APAC)
Sale Manager (EMEA)

Explanation: Returns distinct job titles across all employees.

Q18: Get customer phone numbers

SQL Query:

```
SELECT "customerName", phone FROM customers;
```

Rows Returned: 122

Sample Output:

customerName	phone
Atelier graphique	40.32.2555
Signal Gift Stores	7025551838
Australian Collectors, Co.	03 9520 4555

Explanation: Returns customer name and phone number for all customers.

Q19: Show product MSRP values

SQL Query:

```
SELECT "productName", "MSRP" FROM products;
```

Rows Returned: 110

Sample Output:

productName	MSRP
1969 Harley Davidson Ultimate Chopper	95.70
1952 Alpine Renault 1300	214.30
1996 Moto Guzzi 1100i	118.94

Explanation: Returns product name and MSRP (retail price) for all products.

Q20: List order numbers

SQL Query:

```
SELECT "orderNumber" FROM orders;
```

Rows Returned: 326

Sample Output:

orderNumber

10100

10101

10102

Explanation: Returns just the order number column from the orders table.

Part 1B — JOIN Queries (Q21–Q30)

Q21: Get orders with customer names

SQL Query:

```
SELECT o."orderNumber", o."orderDate", c."customerName"  
      FROM orders o  
      JOIN customers c ON o."customerNumber" = c."customerNumber"
```

Rows Returned: 326

Sample Output:

orderNumber	orderDate	customerName
10100	2003-01-06	Online Diecast Creations Co.
10101	2003-01-09	Blauer See Auto, Co.
10102	2003-01-10	Vitachrome Inc.

Explanation: JOINS orders and customers tables on customerNumber to show order dates with customer names.

Q22: Get employees with office city

SQL Query:

```
SELECT e."firstName", e."lastName", o.city  
      FROM employees e  
      JOIN offices o ON e."officeCode" = o."officeCode";
```

Rows Returned: 23

Sample Output:

firstName	lastName	city
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Diane	Murphy	San Francisco
Mary	Patterson	San Francisco
Jeff	Firrelli	San Francisco

Explanation: JOINS employees and offices tables on officeCode to show which city each employee works in.

Q23: Get payments with customer names

SQL Query:

```
SELECT c."customerName", p.amount, p."paymentDate"  
      FROM payments p  
      JOIN customers c ON p."customerNumber" = c."customerNumber"
```

Rows Returned: 273

Sample Output:

customerName	amount	paymentDate
Atelier graphique	6066.78	2004-10-19
Atelier graphique	14571.44	2003-06-05
Atelier graphique	1676.14	2004-12-18

Explanation: JOINS payments and customers tables to show payment amounts alongside customer names.

Q24: Get order details with product names

SQL Query:

```
SELECT od."orderNumber", p."productName", od."quantityOrdered", od."pri  
      FROM orderdetails od  
      JOIN products p ON od."productCode" = p."productCode";
```

Rows Returned: 2996

Sample Output:

orderNumber	productName	quantityOrdered	priceEach
10100	1917 Grand Touring Sedan	30	136.00
10100	1911 Ford Town Car	50	55.09
10100	1932 Alfa Romeo 8C2300 Spider Sport	22	75.46

Explanation: JOINS orderdetails and products tables to show product names, quantities, and prices per order.

Q25: Get products with product line description

SQL Query:

```
SELECT p."productName", pl."textDescription"
      FROM products p
      JOIN productlines pl ON p."productLine" = pl."productLine";
```

Rows Returned: 110

Sample Output:

productName	textDescription
1969 Harley Davidson Ultimate Chopper	Our motorcycles are state of the art replicas of classic as well as contemporary motorcycle legends such as Harley Davidson, Ducati and Vespa. Models contain stunning details such as official logos, rotating wheels, working kickstand, front suspension, gear-shift lever, footbrake lever, and drive chain. Materials used include diecast and plastic. The models range in size from 1:10 to 1:50 scale and include numerous limited edition and several out-of-production vehicles. All models come fully assembled and ready for display in the home or office. Most include a certificate of authenticity.
1952 Alpine Renault 1300	Attention car enthusiasts: Make your wildest car ownership dreams come true. Whether you are looking for classic muscle cars, dream sports cars or movie-inspired miniatures, you will find great choices in this category. These replicas feature superb attention to detail and craftsmanship and offer features such as working steering system, opening forward compartment, opening rear trunk with removable spare wheel, 4-wheel independent spring suspension, and so on. The models range in size from 1:10 to 1:24 scale and include numerous limited edition and several out-of-production vehicles. All models include a certificate of authenticity from their manufacturers and come fully assembled and ready for display in the home or office.
1996 Moto Guzzi 1100i	Our motorcycles are state of the art replicas of classic as well as contemporary motorcycle legends such as Harley Davidson, Ducati and Vespa. Models contain stunning details such as official logos, rotating wheels, working kickstand, front suspension, gear-shift lever, footbrake lever, and drive chain. Materials used include diecast and plastic. The models range in size from 1:10 to 1:50 scale and include numerous limited edition and several out-of-production vehicles. All models come fully assembled and ready for display in the home or office. Most include a certificate of authenticity.

Explanation: JOINS products and productlines tables to show each product with its category description.

Q26: Get customers with sales rep names

SQL Query:

```
SELECT c."customerName", e."firstName", e."lastName"  
      FROM customers c  
      JOIN employees e ON c."salesRepEmployeeNumber" = e."employeeNumber"
```

Rows Returned: 100

Sample Output:

customerName	firstName	lastName
Atelier graphique	Gerard	Hernandez
Signal Gift Stores	Leslie	Thompson
Australian Collectors, Co.	Andy	Fixter

Explanation: JOINS customers and employees tables to show which sales rep manages each customer.

Q27: Get orders with customer city

SQL Query:

```
SELECT o."orderNumber", o."orderDate", c.city  
      FROM orders o  
      JOIN customers c ON o."customerNumber" = c."customerNumber"
```

Rows Returned: 326

Sample Output:

orderNumber	orderDate	city
10100	2003-01-06	Nashua
10101	2003-01-09	Frankfurt
10102	2003-01-10	NYC

Explanation: JOINS orders and customers to show the city of the customer who placed each order.

Q28: Get employees and their manager

SQL Query:

```
SELECT e."firstName" AS employee, m."firstName" AS manager  
      FROM employees e  
      LEFT JOIN employees m ON e."reportsTo" = m."employeeNumber"
```

Rows Returned: 23

Sample Output:

employee manager

Diane	None
Mary	Diane
Jeff	Diane

Explanation: Self-JOIN on employees table using reportsTo to show each employee paired with their manager.

Q29: Get orderdetails with product vendor

SQL Query:

```
SELECT od."orderNumber", p."productVendor", od."quantityOrdered"
      FROM orderdetails od
      JOIN products p ON od."productCode" = p."productCode";
```

Rows Returned: 2996

Sample Output:

orderNumber	productVendor	quantityOrdered
10100	Welly Diecast Productions	30
10100	Motor City Art Classics	50
10100	Exoto Designs	22

Explanation: JOINS orderdetails and products to show vendor name alongside order quantities.

Q30: Get payments with customer country

SQL Query:

```
SELECT c."customerName", c.country, p.amount
      FROM payments p
      JOIN customers c ON p."customerNumber" = c."customerNumber"
```

Rows Returned: 273

Sample Output:

customerName	country	amount
Atelier graphique	France	6066.78
Atelier graphique	France	14571.44

Atelier graphique France 1676.14

Explanation: JOINS payments and customers to show payment amounts grouped with customer country.

Part 1C — GROUP BY and Aggregation (Q31–Q40)

Q31: Count customers per country

SQL Query:

```
SELECT country, COUNT(*) AS customer_count
      FROM customers
      GROUP BY country
      ORDER BY customer_count DESC;
```

Rows Returned: 28

Sample Output:

country	customer_count
USA	36
Germany	13
France	12

Explanation: Groups customers by country and counts how many are in each, ordered by count descending.

Q32: Total payments per customer

SQL Query:

```
SELECT "customerNumber", SUM(amount) AS total_paid
      FROM payments
      GROUP BY "customerNumber"
      ORDER BY total_paid DESC;
```

Rows Returned: 98

Sample Output:

customerNumber	total_paid
141	715738.98
124	584188.24
114	180585.07

Explanation: Groups payments by customerNumber and sums the total amount paid per customer.

Q33: Number of orders per status

SQL Query:

```
SELECT status, COUNT(*) AS order_count
      FROM orders
     GROUP BY status;
```

Rows Returned: 6

Sample Output:

status	order_count
Shipped	303
In Process	6
Disputed	3

Explanation: Groups orders by status and counts how many orders are in each status category.

Q34: Products per product line

SQL Query:

```
SELECT "productLine", COUNT(*) AS product_count
      FROM products
     GROUP BY "productLine";
```

Rows Returned: 7

Sample Output:

productLine	product_count
Classic Cars	38
Trains	3
Planes	12

Explanation: Groups products by productLine and counts how many products belong to each line.

Q35: Employees per office

SQL Query:

```
SELECT "officeCode", COUNT(*) AS employee_count
      FROM employees
      GROUP BY "officeCode";
```

Rows Returned: 7**Sample Output:**

officeCode	employee_count
6	4
2	2
4	5

Explanation: Groups employees by officeCode and counts staff per office location.

Q36: Total stock per product vendor**SQL Query:**

```
SELECT "productVendor", SUM("quantityInStock") AS total_stock
      FROM products
      GROUP BY "productVendor"
      ORDER BY total_stock DESC;
```

Rows Returned: 13**Sample Output:**

productVendor	total_stock
Gearbox Collectibles	60495
Min Lin Diecast	50089
Classic Metal Creations	45408

Explanation: Groups products by vendor and sums total quantity in stock per vendor.

Q37: Average buy price per product line**SQL Query:**

```
SELECT "productLine", ROUND(AVG("buyPrice"), 2) AS avg_buy_price
      FROM products
      GROUP BY "productLine";
```

Rows Returned: 7

Sample Output:

productLine avg_buy_price

Classic Cars	64.45
Trains	43.92
Planes	49.63

Explanation: Groups products by product line and calculates the average buy price for each.

Q38: Orders per customer

SQL Query:

```
SELECT "customerNumber", COUNT(*) AS order_count
      FROM orders
      GROUP BY "customerNumber"
      ORDER BY order_count DESC;
```

Rows Returned: 98

Sample Output:

customerNumber order_count

141	26
124	17
148	5

Explanation: Groups orders by customerNumber and counts how many orders each customer has placed.

Q39: Max MSRP per product line

SQL Query:

```
SELECT "productLine", MAX("MSRP") AS max_msrp
      FROM products
      GROUP BY "productLine";
```

Rows Returned: 7

Sample Output:

productLine max_msrp

Classic Cars	214.30
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Trains	100.84
Planes	157.69

Explanation: Groups products by product line and finds the highest MSRP in each category.

Q40: Min buy price per vendor

SQL Query:

```
SELECT "productVendor", MIN("buyPrice") AS min_buy_price
      FROM products
      GROUP BY "productVendor";
```

Rows Returned: 13

Sample Output:

productVendor	min_buy_price
Welly Diecast Productions	24.23
Motor City Art Classics	22.57
Classic Metal Creations	20.61

Explanation: Groups products by vendor and finds the lowest buy price offered by each vendor.

Part 1D — Aggregate Functions (Q41–Q50)

Q41: Total number of customers

SQL Query:

```
SELECT COUNT(*) AS total_customers FROM customers;
```

Rows Returned: 1

Sample Output:

total_customers
122

Explanation: Counts the total number of customer records in the customers table.

Q42: Total number of products

SQL Query:

```
SELECT COUNT(*) AS total_products FROM products;
```

Rows Returned: 1**Sample Output:**

total_products
110

Explanation: Counts the total number of products in the products table.

Q43: Total revenue from payments**SQL Query:**

```
SELECT SUM(amount) AS total_revenue FROM payments;
```

Rows Returned: 1**Sample Output:**

total_revenue
8853839.23

Explanation: Sums all payment amounts to get total revenue of \$8,853,839.23.

Q44: Average product price**SQL Query:**

```
SELECT ROUND(AVG("buyPrice"), 2) AS avg_buy_price FROM products;
```

Rows Returned: 1**Sample Output:**

avg_buy_price
54.40

Explanation: Calculates the average buy price across all products.

Q45: Max payment amount

SQL Query:

```
SELECT MAX(amount) AS max_payment FROM payments;
```

Rows Returned: 1**Sample Output:**

max_payment
120166.58

Explanation: Finds the single largest payment amount recorded (\$120,166.58).

Q46: Min payment amount**SQL Query:**

```
SELECT MIN(amount) AS min_payment FROM payments;
```

Rows Returned: 1**Sample Output:**

min_payment
615.45

Explanation: Finds the smallest payment amount recorded (\$615.45).

Q47: Count total orders**SQL Query:**

```
SELECT COUNT(*) AS total_orders FROM orders;
```

Rows Returned: 1**Sample Output:**

total_orders
326

Explanation: Counts the total number of orders placed (326 orders).

Q48: Total quantity in stock

SQL Query:

```
SELECT SUM("quantityInStock") AS total_stock FROM products;
```

Rows Returned: 1**Sample Output:**

total_stock
555131

Explanation: Sums quantityInStock across all products to get total inventory.

Q49: Average MSRP**SQL Query:**

```
SELECT ROUND(AVG("MSRP"), 2) AS avg_msrp FROM products;
```

Rows Returned: 1**Sample Output:**

avg_msrp
100.44

Explanation: Calculates the average MSRP (retail price) across all products.

Q50: Number of employees**SQL Query:**

```
SELECT COUNT(*) AS total_employees FROM employees;
```

Rows Returned: 1**Sample Output:**

total_employees
23

Explanation: Counts the total number of employees in the company (23 employees).

Summary

Metric	Value
Total Questions	50
Successful Queries	50
Failed Queries	0
Success Rate	100%